

Trailing -> Forward correlation call-outs (|r| >= 0.5)

Strong = |r| >= 0.70 | Moderate = 0.50 <= |r| < 0.70. Cells ranked by |r|.

avg() = mean of that factor across the days passing the panel filter.

Small-n panels (n < ~40) are noise-dominated; weight accordingly.

Forward 1-Day portfolio return (filtered by book_fwd_1d)

UP > +2.0% (n=296)

MODERATE (0.50-0.70):

-0.62	spy_vol_trl_5d	avg=+16.7%	->	psq_fwd_1d	avg=-1.7%
-0.61	qqq_vol_trl_5d	avg=+21.6%	->	psq_fwd_1d	avg=-1.7%
-0.58	qqq_vol_trl_20d	avg=+22.8%	->	psq_fwd_1d	avg=-1.7%
+0.58	spy_trl_20d	avg=+0.3%	->	psq_fwd_1d	avg=-1.7%
+0.57	qqq_vol_trl_20d	avg=+22.8%	->	universe_fwd_1d	avg=+2.3%
-0.57	spy_vol_trl_20d	avg=+17.8%	->	psq_fwd_1d	avg=-1.7%
+0.55	qqq_vol_trl_5d	avg=+21.6%	->	universe_fwd_1d	avg=+2.3%
+0.55	spy_vol_trl_5d	avg=+16.7%	->	universe_fwd_1d	avg=+2.3%
+0.54	spy_vol_trl_20d	avg=+17.8%	->	universe_fwd_1d	avg=+2.3%
+0.54	qqq_trl_20d	avg=+0.6%	->	psq_fwd_1d	avg=-1.7%
-0.54	spy_trl_20d	avg=+0.3%	->	universe_fwd_1d	avg=+2.3%
-0.54	universe_trl_20d	avg=+1.6%	->	bottom10_fwd_1d	avg=+2.4%
+0.53	universe_trl_20d	avg=+1.6%	->	psq_fwd_1d	avg=-1.7%
-0.53	psq_trl_20d	avg=-0.4%	->	psq_fwd_1d	avg=-1.7%
-0.53	qqq_trl_20d	avg=+0.6%	->	bottom10_fwd_1d	avg=+2.4%
-0.53	book_trl_20d	avg=+2.8%	->	bottom10_fwd_1d	avg=+2.4%
-0.53	qqq_trl_20d	avg=+0.6%	->	universe_fwd_1d	avg=+2.3%
-0.53	universe_trl_20d	avg=+1.6%	->	universe_fwd_1d	avg=+2.3%
-0.52	qqq_vol_trl_5d	avg=+21.6%	->	soxs_fwd_1d	avg=-7.7%
+0.52	psq_trl_20d	avg=-0.4%	->	universe_fwd_1d	avg=+2.3%
+0.52	book_trl_20d	avg=+2.8%	->	psq_fwd_1d	avg=-1.7%
+0.51	psq_trl_20d	avg=-0.4%	->	bottom10_fwd_1d	avg=+2.4%
+0.51	qqq_vol_trl_20d	avg=+22.8%	->	bottom10_fwd_1d	avg=+2.4%
-0.51	spy_vol_trl_5d	avg=+16.7%	->	soxs_fwd_1d	avg=-7.7%
-0.50	book_trl_20d	avg=+2.8%	->	universe_fwd_1d	avg=+2.3%
+0.50	qqq_vol_trl_5d	avg=+21.6%	->	smh_fwd_1d	avg=+2.5%

UP > +3.5% (n=68)

MODERATE (0.50-0.70):

-0.70	book_trl_20d	avg=+2.5%	->	bottom10_fwd_1d	avg=+3.1%
-0.67	qqq_trl_20d	avg=-0.3%	->	bottom10_fwd_1d	avg=+3.1%
-0.67	qqq_vol_trl_20d	avg=+25.1%	->	psq_fwd_1d	avg=-2.6%
+0.66	book_trl_20d	avg=+2.5%	->	spread_fwd_1d	avg=+0.5%
-0.66	universe_trl_20d	avg=+0.7%	->	bottom10_fwd_1d	avg=+3.1%
+0.66	spy_trl_20d	avg=-1.1%	->	psq_fwd_1d	avg=-2.6%
-0.66	spy_trl_20d	avg=-1.1%	->	universe_fwd_1d	avg=+3.1%
-0.65	qqq_trl_20d	avg=-0.3%	->	universe_fwd_1d	avg=+3.1%
-0.65	book_trl_20d	avg=+2.5%	->	universe_fwd_1d	avg=+3.1%
-0.65	universe_trl_20d	avg=+0.7%	->	universe_fwd_1d	avg=+3.1%
+0.65	spy_trl_20d	avg=-1.1%	->	spread_fwd_20d	avg=-1.3%
-0.64	spy_vol_trl_20d	avg=+20.6%	->	psq_fwd_1d	avg=-2.6%
+0.64	psq_trl_20d	avg=+0.6%	->	bottom10_fwd_1d	avg=+3.1%
-0.64	bottom10_trl_20d	avg=+0.7%	->	bottom10_fwd_1d	avg=+3.1%
+0.64	psq_trl_20d	avg=+0.6%	->	universe_fwd_1d	avg=+3.1%
-0.63	spy_trl_20d	avg=-1.1%	->	bottom10_fwd_1d	avg=+3.1%
-0.63	bottom10_trl_20d	avg=+0.7%	->	universe_fwd_1d	avg=+3.1%
+0.63	universe_trl_20d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
+0.63	book_trl_20d	avg=+2.5%	->	spread_fwd_20d	avg=-1.3%
+0.63	qqq_trl_20d	avg=-0.3%	->	psq_fwd_1d	avg=-2.6%
+0.63	qqq_vol_trl_20d	avg=+25.1%	->	universe_fwd_1d	avg=+3.1%
-0.62	spy_vol_trl_5d	avg=+21.8%	->	psq_fwd_1d	avg=-2.6%

+0.62	qqq_vol_trl_20d	avg=+25.1%	->	bottom10_fwd_1d	avg=+3.1%
+0.62	book_trl_20d	avg=+2.5%	->	psq_fwd_1d	avg=-2.6%
-0.61	qqq_vol_trl_5d	avg=+26.4%	->	psq_fwd_1d	avg=-2.6%
+0.61	spy_vol_trl_20d	avg=+20.6%	->	universe_fwd_1d	avg=+3.1%
-0.61	psq_trl_20d	avg=+0.6%	->	psq_fwd_1d	avg=-2.6%
+0.60	qqq_trl_20d	avg=-0.3%	->	spread_fwd_20d	avg=-1.3%
-0.60	qqq_vol_trl_5d	avg=+26.4%	->	spread_fwd_20d	avg=-1.3%
+0.60	spy_vol_trl_20d	avg=+20.6%	->	bottom10_fwd_1d	avg=+3.1%
+0.60	qqq_vol_trl_20d	avg=+25.1%	->	bottom10_fwd_20d	avg=+8.2%
-0.60	top10_trl_20d	avg=+2.4%	->	bottom10_fwd_1d	avg=+3.1%
-0.60	spy_vol_trl_5d	avg=+21.8%	->	spread_fwd_20d	avg=-1.3%
+0.59	spy_vol_trl_20d	avg=+20.6%	->	bottom10_fwd_20d	avg=+8.2%
-0.58	smh_trl_20d	avg=+1.8%	->	bottom10_fwd_1d	avg=+3.1%
+0.58	bottom10_trl_20d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
-0.58	psq_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=-1.3%
+0.58	qqq_trl_20d	avg=-0.3%	->	spread_fwd_1d	avg=+0.5%
+0.57	spy_trl_5d	avg=-0.7%	->	psq_fwd_1d	avg=-2.6%
-0.57	qqq_vol_trl_20d	avg=+25.1%	->	spread_fwd_20d	avg=-1.3%
+0.57	universe_trl_20d	avg=+0.7%	->	spread_fwd_1d	avg=+0.5%
+0.57	spy_vol_trl_5d	avg=+21.8%	->	universe_fwd_1d	avg=+3.1%
-0.57	smh_trl_20d	avg=+1.8%	->	universe_fwd_1d	avg=+3.1%
-0.57	top10_trl_20d	avg=+2.4%	->	universe_fwd_1d	avg=+3.1%
+0.56	bottom10_trl_20d	avg=+0.7%	->	spread_fwd_20d	avg=-1.3%
-0.56	spy_vol_trl_20d	avg=+20.6%	->	spread_fwd_20d	avg=-1.3%
+0.56	qqq_vol_trl_5d	avg=+26.4%	->	universe_fwd_1d	avg=+3.1%
+0.56	book_trl_5d	avg=-0.1%	->	spread_fwd_1d	avg=+0.5%
+0.55	smh_trl_20d	avg=+1.8%	->	spread_fwd_20d	avg=-1.3%
+0.55	universe_trl_20d	avg=+0.7%	->	spread_fwd_20d	avg=-1.3%
+0.55	bottom10_trl_20d	avg=+0.7%	->	spread_fwd_1d	avg=+0.5%
-0.54	qqq_vol_trl_20d	avg=+25.1%	->	psq_fwd_20d	avg=-4.7%
+0.54	qqq_vol_trl_20d	avg=+25.1%	->	smh_fwd_1d	avg=+3.7%
-0.54	qqq_vol_trl_20d	avg=+25.1%	->	soxs_fwd_1d	avg=-11.6%
-0.54	spy_vol_trl_20d	avg=+20.6%	->	psq_fwd_20d	avg=-4.7%
-0.54	spy_vol_trl_1d	avg=+19.4%	->	spread_fwd_20d	avg=-1.3%
-0.54	psq_trl_20d	avg=+0.6%	->	spread_fwd_1d	avg=+0.5%
+0.54	book_trl_5d	avg=-0.1%	->	psq_fwd_1d	avg=-2.6%
-0.53	qqq_vol_trl_5d	avg=+26.4%	->	soxs_fwd_1d	avg=-11.6%
+0.53	smh_trl_20d	avg=+1.8%	->	psq_fwd_1d	avg=-2.6%
+0.53	top10_trl_20d	avg=+2.4%	->	psq_fwd_1d	avg=-2.6%
+0.52	spy_vol_trl_20d	avg=+20.6%	->	smh_fwd_1d	avg=+3.7%
-0.52	spy_trl_20d	avg=-1.1%	->	smh_fwd_1d	avg=+3.7%
+0.52	top10_trl_20d	avg=+2.4%	->	spread_fwd_1d	avg=+0.5%
-0.52	spy_vol_trl_5d	avg=+21.8%	->	soxs_fwd_1d	avg=-11.6%
+0.52	spy_trl_20d	avg=-1.1%	->	soxs_fwd_1d	avg=-11.6%
+0.52	smh_trl_20d	avg=+1.8%	->	spread_fwd_1d	avg=+0.5%
-0.52	spy_vol_trl_20d	avg=+20.6%	->	soxs_fwd_1d	avg=-11.6%
-0.52	spy_trl_5d	avg=-0.7%	->	universe_fwd_1d	avg=+3.1%
-0.52	score_top10_trl_20d	avg=0.840	->	universe_fwd_1d	avg=+3.1%
+0.52	spy_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_1d	avg=+3.1%
+0.52	qqq_vol_trl_5d	avg=+26.4%	->	bottom10_fwd_20d	avg=+8.2%
+0.52	spy_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_20d	avg=+8.2%
+0.51	qqq_trl_5d	avg=-0.6%	->	psq_fwd_1d	avg=-2.6%
-0.51	psq_trl_5d	avg=+0.7%	->	psq_fwd_1d	avg=-2.6%
-0.51	book_trl_5d	avg=-0.1%	->	bottom10_fwd_1d	avg=+3.1%
+0.51	qqq_vol_trl_5d	avg=+26.4%	->	bottom10_fwd_1d	avg=+3.1%
+0.50	spy_trl_20d	avg=-1.1%	->	spread_fwd_1d	avg=+0.5%
+0.50	spy_vol_trl_5d	avg=+21.8%	->	smh_fwd_1d	avg=+3.7%
+0.50	qqq_vol_trl_5d	avg=+26.4%	->	smh_fwd_1d	avg=+3.7%

-0.50	score_top10_trl_20d	avg=0.840	->	bottom10_fwd_1d	avg=+3.1%
-0.50	universe_trl_20d	avg=+0.7%	->	smh_fwd_1d	avg=+3.7%

DOWN < -2.0% (n=250)

MODERATE (0.50-0.70):

+0.53	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_1d	avg=+2.1%
+0.51	top10_trl_20d	avg=+2.2%	->	bottom10_fwd_1d	avg=-2.7%
-0.51	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_1d	avg=-2.7%
+0.51	qqq_vol_trl_5d	avg=+22.9%	->	psq_fwd_1d	avg=+2.1%
+0.51	book_trl_20d	avg=+2.2%	->	bottom10_fwd_1d	avg=-2.7%

DOWN < -3.5% (n=77)

MODERATE (0.50-0.70):

+0.63	spy_vol_trl_5d	avg=+21.9%	->	psq_fwd_1d	avg=+3.1%
+0.62	spy_vol_trl_1d	avg=+17.4%	->	psq_fwd_1d	avg=+3.1%
-0.62	qqq_vol_trl_5d	avg=+26.8%	->	spread_fwd_20d	avg=-2.9%
+0.60	qqq_vol_trl_5d	avg=+26.8%	->	psq_fwd_1d	avg=+3.1%
-0.59	spy_vol_trl_5d	avg=+21.9%	->	spread_fwd_20d	avg=-2.9%
-0.58	spy_vol_trl_1d	avg=+17.4%	->	universe_fwd_1d	avg=-3.8%
-0.58	qqq_vol_trl_20d	avg=+24.0%	->	spread_fwd_20d	avg=-2.9%
+0.57	qqq_vol_trl_1d	avg=+21.2%	->	psq_fwd_1d	avg=+3.1%
-0.54	spy_vol_trl_20d	avg=+19.1%	->	spread_fwd_20d	avg=-2.9%
+0.53	spy_vol_trl_20d	avg=+19.1%	->	psq_fwd_1d	avg=+3.1%
-0.53	spy_vol_trl_5d	avg=+21.9%	->	universe_fwd_1d	avg=-3.8%
-0.53	qqq_vol_trl_1d	avg=+21.2%	->	universe_fwd_1d	avg=-3.8%
+0.52	spy_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_20d	avg=+1.4%
+0.52	spy_vol_trl_1d	avg=+17.4%	->	soxs_fwd_1d	avg=+13.7%
+0.52	qqq_vol_trl_20d	avg=+24.0%	->	psq_fwd_1d	avg=+3.1%
+0.52	spy_trl_20d	avg=-1.0%	->	spread_fwd_20d	avg=-2.9%
-0.52	spy_vol_trl_1d	avg=+17.4%	->	smh_fwd_1d	avg=-4.4%
+0.51	spy_vol_trl_20d	avg=+19.1%	->	bottom10_fwd_20d	avg=+1.4%
-0.50	smh_trl_20d	avg=+0.9%	->	psq_fwd_1d	avg=+3.1%
+0.50	book_trl_20d	avg=+1.1%	->	bottom10_fwd_1d	avg=-4.0%
+0.50	qqq_vol_trl_5d	avg=+26.8%	->	bottom10_fwd_20d	avg=+1.4%

Forward 5-Day portfolio return (filtered by book_fwd_5d)

UP > +4.5% (n=340)

MODERATE (0.50-0.70):

-0.58	qqq_vol_trl_20d	avg=+23.2%	->	psq_fwd_5d	avg=-3.4%
+0.58	qqq_vol_trl_20d	avg=+23.2%	->	bottom10_fwd_5d	avg=+5.1%
+0.57	qqq_vol_trl_20d	avg=+23.2%	->	universe_fwd_5d	avg=+5.0%
-0.56	spy_vol_trl_20d	avg=+18.4%	->	psq_fwd_5d	avg=-3.4%
+0.56	spy_vol_trl_20d	avg=+18.4%	->	bottom10_fwd_20d	avg=+7.9%
+0.55	qqq_vol_trl_20d	avg=+23.2%	->	bottom10_fwd_20d	avg=+7.9%
+0.55	spy_vol_trl_20d	avg=+18.4%	->	bottom10_fwd_5d	avg=+5.1%
-0.54	qqq_vol_trl_5d	avg=+21.8%	->	psq_fwd_5d	avg=-3.4%
+0.54	spy_vol_trl_5d	avg=+16.9%	->	bottom10_fwd_20d	avg=+7.9%
+0.54	spy_vol_trl_20d	avg=+18.4%	->	universe_fwd_5d	avg=+5.0%
+0.54	book_trl_20d	avg=+2.8%	->	spread_fwd_5d	avg=+1.0%
-0.53	spy_vol_trl_5d	avg=+16.9%	->	psq_fwd_5d	avg=-3.4%
-0.53	book_trl_20d	avg=+2.8%	->	bottom10_fwd_5d	avg=+5.1%
+0.52	universe_trl_20d	avg=+1.9%	->	spread_fwd_5d	avg=+1.0%
+0.52	spy_vol_trl_5d	avg=+16.9%	->	universe_fwd_5d	avg=+5.0%
+0.52	qqq_vol_trl_5d	avg=+21.8%	->	universe_fwd_5d	avg=+5.0%
-0.52	universe_trl_20d	avg=+1.9%	->	bottom10_fwd_5d	avg=+5.1%
+0.52	qqq_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_20d	avg=+7.9%
+0.52	spy_vol_trl_5d	avg=+16.9%	->	bottom10_fwd_5d	avg=+5.1%
+0.51	qqq_vol_trl_5d	avg=+21.8%	->	bottom10_fwd_5d	avg=+5.1%
+0.51	qqq_trl_20d	avg=+0.8%	->	spread_fwd_5d	avg=+1.0%

UP > +6.5% (n=127)

MODERATE (0.50-0.70):

+0.66	book_trl_20d	avg=+3.9%	->	spread_fwd_5d	avg=+2.0%
-0.65	book_trl_20d	avg=+3.9%	->	bottom10_fwd_5d	avg=+5.9%
-0.65	qqq_vol_trl_5d	avg=+21.9%	->	psq_fwd_5d	avg=-3.9%
+0.65	spy_vol_trl_5d	avg=+17.7%	->	bottom10_fwd_20d	avg=+9.1%
+0.64	qqq_vol_trl_20d	avg=+25.3%	->	bottom10_fwd_20d	avg=+9.1%
+0.64	spy_vol_trl_20d	avg=+20.7%	->	bottom10_fwd_20d	avg=+9.1%
+0.63	qqq_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_20d	avg=+9.1%
-0.62	qqq_vol_trl_20d	avg=+25.3%	->	psq_fwd_5d	avg=-3.9%
-0.62	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_5d	avg=-3.9%
-0.61	universe_trl_20d	avg=+2.8%	->	bottom10_fwd_5d	avg=+5.9%
+0.60	universe_trl_20d	avg=+2.8%	->	spread_fwd_5d	avg=+2.0%
+0.60	qqq_vol_trl_5d	avg=+21.9%	->	bottom10_fwd_5d	avg=+5.9%
-0.60	spy_vol_trl_20d	avg=+20.7%	->	psq_fwd_5d	avg=-3.9%
+0.60	qqq_vol_trl_20d	avg=+25.3%	->	bottom10_fwd_5d	avg=+5.9%
+0.59	spy_vol_trl_5d	avg=+17.7%	->	bottom10_fwd_5d	avg=+5.9%
+0.59	qqq_trl_20d	avg=+1.6%	->	spread_fwd_5d	avg=+2.0%
-0.58	bottom10_trl_20d	avg=+3.2%	->	bottom10_fwd_5d	avg=+5.9%
+0.58	bottom10_trl_20d	avg=+3.2%	->	spread_fwd_5d	avg=+2.0%
+0.57	qqq_vol_trl_5d	avg=+21.9%	->	universe_fwd_5d	avg=+5.8%
+0.57	spy_trl_20d	avg=+0.9%	->	spread_fwd_5d	avg=+2.0%
+0.57	spy_vol_trl_20d	avg=+20.7%	->	bottom10_fwd_5d	avg=+5.9%
-0.57	qqq_trl_20d	avg=+1.6%	->	bottom10_fwd_5d	avg=+5.9%
-0.56	spy_vol_trl_20d	avg=+20.7%	->	psq_fwd_20d	avg=-5.1%
-0.56	qqq_vol_trl_20d	avg=+25.3%	->	spread_fwd_5d	avg=+2.0%
-0.56	qqq_vol_trl_20d	avg=+25.3%	->	psq_fwd_20d	avg=-5.1%
-0.56	spy_trl_20d	avg=+0.9%	->	bottom10_fwd_5d	avg=+5.9%
+0.55	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_5d	avg=+5.8%
+0.55	top10_trl_20d	avg=+4.4%	->	spread_fwd_5d	avg=+2.0%
-0.55	spy_trl_20d	avg=+0.9%	->	bottom10_fwd_20d	avg=+9.1%
+0.55	spy_trl_20d	avg=+0.9%	->	spread_fwd_20d	avg=+1.8%
+0.55	book_trl_20d	avg=+3.9%	->	spread_fwd_20d	avg=+1.8%
-0.55	qqq_vol_trl_5d	avg=+21.9%	->	spread_fwd_5d	avg=+2.0%
-0.55	spy_vol_trl_5d	avg=+17.7%	->	psq_fwd_20d	avg=-5.1%
-0.55	book_trl_20d	avg=+3.9%	->	universe_fwd_5d	avg=+5.8%
+0.54	book_trl_20d	avg=+3.9%	->	psq_fwd_5d	avg=-3.9%
-0.54	spy_vol_trl_20d	avg=+20.7%	->	spread_fwd_5d	avg=+2.0%
+0.54	qqq_vol_trl_20d	avg=+25.3%	->	universe_fwd_5d	avg=+5.8%
-0.54	spy_vol_trl_5d	avg=+17.7%	->	spread_fwd_5d	avg=+2.0%
-0.53	book_trl_20d	avg=+3.9%	->	bottom10_fwd_20d	avg=+9.1%
+0.53	spy_trl_20d	avg=+0.9%	->	psq_fwd_5d	avg=-3.9%
-0.53	psq_trl_20d	avg=-1.5%	->	spread_fwd_5d	avg=+2.0%
+0.53	qqq_trl_20d	avg=+1.6%	->	spread_fwd_20d	avg=+1.8%
-0.53	qqq_vol_trl_5d	avg=+21.9%	->	psq_fwd_20d	avg=-5.1%
+0.53	universe_trl_20d	avg=+2.8%	->	psq_fwd_5d	avg=-3.9%
+0.52	smh_trl_20d	avg=+3.5%	->	spread_fwd_5d	avg=+2.0%
-0.52	universe_trl_20d	avg=+2.8%	->	universe_fwd_5d	avg=+5.8%
+0.52	smh_trl_20d	avg=+3.5%	->	spread_fwd_20d	avg=+1.8%
-0.52	spy_trl_20d	avg=+0.9%	->	universe_fwd_5d	avg=+5.8%
-0.51	top10_trl_20d	avg=+4.4%	->	bottom10_fwd_5d	avg=+5.9%
+0.51	spy_vol_trl_20d	avg=+20.7%	->	universe_fwd_5d	avg=+5.8%
+0.51	psq_trl_20d	avg=-1.5%	->	bottom10_fwd_5d	avg=+5.9%
+0.50	spy_vol_trl_5d	avg=+17.7%	->	universe_fwd_20d	avg=+8.3%

DOWN < -4.5% (n=222)

MODERATE (0.50-0.70):

-0.57	qqq_vol_trl_5d	avg=+22.1%	->	bottom10_fwd_5d	avg=-6.2%
-0.54	qqq_vol_trl_5d	avg=+22.1%	->	universe_fwd_5d	avg=-5.8%
-0.54	spy_vol_trl_5d	avg=+17.0%	->	bottom10_fwd_5d	avg=-6.2%

-0.53	qqq_vol_trl_20d	avg=+21.0%	->	bottom10_fwd_5d	avg=-6.2%
-0.53	qqq_vol_trl_20d	avg=+21.0%	->	universe_fwd_5d	avg=-5.8%
-0.52	spy_vol_trl_5d	avg=+17.0%	->	universe_fwd_5d	avg=-5.8%
-0.51	spy_vol_trl_20d	avg=+15.5%	->	universe_fwd_5d	avg=-5.8%
-0.51	spy_vol_trl_20d	avg=+15.5%	->	bottom10_fwd_5d	avg=-6.2%

DOWN < -6.5% (n=103)

MODERATE (0.50-0.70):

-0.66	qqq_vol_trl_20d	avg=+22.4%	->	universe_fwd_5d	avg=-7.8%
-0.66	qqq_vol_trl_20d	avg=+22.4%	->	bottom10_fwd_5d	avg=-8.5%
-0.65	spy_vol_trl_20d	avg=+16.8%	->	universe_fwd_5d	avg=-7.8%
-0.65	qqq_vol_trl_5d	avg=+24.5%	->	bottom10_fwd_5d	avg=-8.5%
-0.64	spy_vol_trl_20d	avg=+16.8%	->	bottom10_fwd_5d	avg=-8.5%
-0.62	qqq_vol_trl_5d	avg=+24.5%	->	universe_fwd_5d	avg=-7.8%
-0.62	spy_vol_trl_5d	avg=+19.5%	->	bottom10_fwd_5d	avg=-8.5%
-0.60	qqq_vol_trl_20d	avg=+22.4%	->	top10_fwd_5d	avg=-8.1%
-0.60	spy_vol_trl_5d	avg=+19.5%	->	universe_fwd_5d	avg=-7.8%
-0.56	spy_vol_trl_20d	avg=+16.8%	->	top10_fwd_5d	avg=-8.1%
-0.56	spy_vol_trl_1d	avg=+18.2%	->	bottom10_fwd_5d	avg=-8.5%
+0.55	book_trl_5d	avg=+0.2%	->	spread_fwd_20d	avg=-3.6%
-0.55	spy_vol_trl_20d	avg=+16.8%	->	smh_fwd_5d	avg=-8.1%
-0.54	spy_vol_trl_1d	avg=+18.2%	->	universe_fwd_5d	avg=-7.8%
-0.53	qqq_vol_trl_5d	avg=+24.5%	->	bottom10_fwd_1d	avg=-1.9%
+0.53	spy_trl_5d	avg=-0.3%	->	spread_fwd_20d	avg=-3.6%
-0.53	spy_vol_trl_5d	avg=+19.5%	->	bottom10_fwd_1d	avg=-1.9%
-0.53	qqq_vol_trl_20d	avg=+22.4%	->	smh_fwd_5d	avg=-8.1%
-0.52	qqq_vol_trl_1d	avg=+22.2%	->	bottom10_fwd_5d	avg=-8.5%
-0.52	spy_vol_trl_5d	avg=+19.5%	->	smh_fwd_5d	avg=-8.1%
-0.51	qqq_vol_trl_5d	avg=+24.5%	->	smh_fwd_5d	avg=-8.1%

Forward 20-Day portfolio return (filtered by book_fwd_20d)

UP > +7.5% (n=655)

MODERATE (0.50-0.70):

+0.68	qqq_vol_trl_20d	avg=+21.6%	->	bottom10_fwd_20d	avg=+9.9%
+0.67	spy_vol_trl_20d	avg=+17.5%	->	bottom10_fwd_20d	avg=+9.9%
+0.65	spy_vol_trl_5d	avg=+16.5%	->	bottom10_fwd_20d	avg=+9.9%
+0.64	qqq_vol_trl_5d	avg=+20.3%	->	bottom10_fwd_20d	avg=+9.9%
-0.61	spy_vol_trl_20d	avg=+17.5%	->	psq_fwd_20d	avg=-5.7%
-0.60	spy_vol_trl_5d	avg=+16.5%	->	psq_fwd_20d	avg=-5.7%
-0.59	qqq_vol_trl_20d	avg=+21.6%	->	psq_fwd_20d	avg=-5.7%
+0.59	qqq_vol_trl_20d	avg=+21.6%	->	universe_fwd_20d	avg=+9.4%
+0.58	spy_vol_trl_20d	avg=+17.5%	->	universe_fwd_20d	avg=+9.4%
+0.57	spy_vol_trl_5d	avg=+16.5%	->	universe_fwd_20d	avg=+9.4%
+0.57	qqq_vol_trl_5d	avg=+20.3%	->	universe_fwd_20d	avg=+9.4%
-0.56	qqq_vol_trl_5d	avg=+20.3%	->	psq_fwd_20d	avg=-5.7%

UP > +12.5% (n=268)

STRONG (|r|>=0.70):

+0.76	spy_vol_trl_5d	avg=+19.7%	->	bottom10_fwd_20d	avg=+12.2%
+0.75	qqq_vol_trl_5d	avg=+24.0%	->	bottom10_fwd_20d	avg=+12.2%
+0.75	qqq_vol_trl_20d	avg=+24.6%	->	bottom10_fwd_20d	avg=+12.2%
+0.74	spy_vol_trl_20d	avg=+20.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.73	spy_trl_20d	avg=+0.2%	->	bottom10_fwd_20d	avg=+12.2%

MODERATE (0.50-0.70):

-0.66	spy_vol_trl_20d	avg=+20.2%	->	psq_fwd_20d	avg=-7.0%
+0.66	spy_trl_20d	avg=+0.2%	->	psq_fwd_20d	avg=-7.0%
-0.66	universe_trl_20d	avg=+1.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.65	qqq_vol_trl_20d	avg=+24.6%	->	psq_fwd_20d	avg=-7.0%
-0.65	spy_vol_trl_5d	avg=+19.7%	->	psq_fwd_20d	avg=-7.0%
+0.65	qqq_vol_trl_5d	avg=+24.0%	->	universe_fwd_20d	avg=+11.9%

-0.65	bottom10_trl_20d	avg=+1.7%	->	bottom10_fwd_20d	avg=+12.2%
+0.65	spy_vol_trl_5d	avg=+19.7%	->	universe_fwd_20d	avg=+11.9%
+0.63	universe_trl_20d	avg=+1.2%	->	psq_fwd_20d	avg=-7.0%
-0.62	book_trl_20d	avg=+1.8%	->	bottom10_fwd_20d	avg=+12.2%
+0.62	spy_trl_20d	avg=+0.2%	->	spread_fwd_20d	avg=+3.6%
+0.62	qqq_vol_trl_20d	avg=+24.6%	->	universe_fwd_20d	avg=+11.9%
-0.62	qqq_vol_trl_5d	avg=+24.0%	->	psq_fwd_20d	avg=-7.0%
+0.62	spy_vol_trl_20d	avg=+20.2%	->	universe_fwd_20d	avg=+11.9%
-0.61	qqq_trl_20d	avg=+0.6%	->	bottom10_fwd_20d	avg=+12.2%
-0.60	spy_vol_trl_5d	avg=+19.7%	->	spread_fwd_20d	avg=+3.6%
+0.59	book_trl_20d	avg=+1.8%	->	spread_fwd_20d	avg=+3.6%
-0.59	spy_trl_20d	avg=+0.2%	->	universe_fwd_20d	avg=+11.9%
+0.59	qqq_trl_20d	avg=+0.6%	->	psq_fwd_20d	avg=-7.0%
-0.59	qqq_vol_trl_20d	avg=+24.6%	->	spread_fwd_20d	avg=+3.6%
-0.59	qqq_vol_trl_5d	avg=+24.0%	->	spread_fwd_20d	avg=+3.6%
+0.58	psq_trl_20d	avg=-0.4%	->	bottom10_fwd_20d	avg=+12.2%
-0.58	spy_vol_trl_20d	avg=+20.2%	->	spread_fwd_20d	avg=+3.6%
-0.57	universe_trl_20d	avg=+1.2%	->	universe_fwd_20d	avg=+11.9%
+0.57	top10_trl_20d	avg=+1.5%	->	spread_fwd_20d	avg=+3.6%
-0.56	psq_trl_20d	avg=-0.4%	->	psq_fwd_20d	avg=-7.0%
+0.55	bottom10_trl_20d	avg=+1.7%	->	psq_fwd_20d	avg=-7.0%
+0.55	qqq_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=+3.6%
+0.54	spy_vol_trl_1d	avg=+15.2%	->	bottom10_fwd_20d	avg=+12.2%
-0.53	qqq_trl_20d	avg=+0.6%	->	universe_fwd_20d	avg=+11.9%
+0.53	book_trl_20d	avg=+1.8%	->	psq_fwd_20d	avg=-7.0%
-0.52	bottom10_trl_20d	avg=+1.7%	->	universe_fwd_20d	avg=+11.9%
+0.52	spy_trl_20d	avg=+0.2%	->	spread_fwd_5d	avg=+0.4%
-0.52	spy_trl_20d	avg=+0.2%	->	bottom10_fwd_5d	avg=+3.6%
-0.51	psq_trl_20d	avg=-0.4%	->	spread_fwd_20d	avg=+3.6%
+0.51	smh_trl_20d	avg=+1.4%	->	spread_fwd_20d	avg=+3.6%
+0.51	universe_trl_20d	avg=+1.2%	->	spread_fwd_20d	avg=+3.6%
+0.50	psq_trl_20d	avg=-0.4%	->	universe_fwd_20d	avg=+11.9%
-0.50	book_trl_20d	avg=+1.8%	->	universe_fwd_20d	avg=+11.9%

DOWN < -7.5% (n=215)

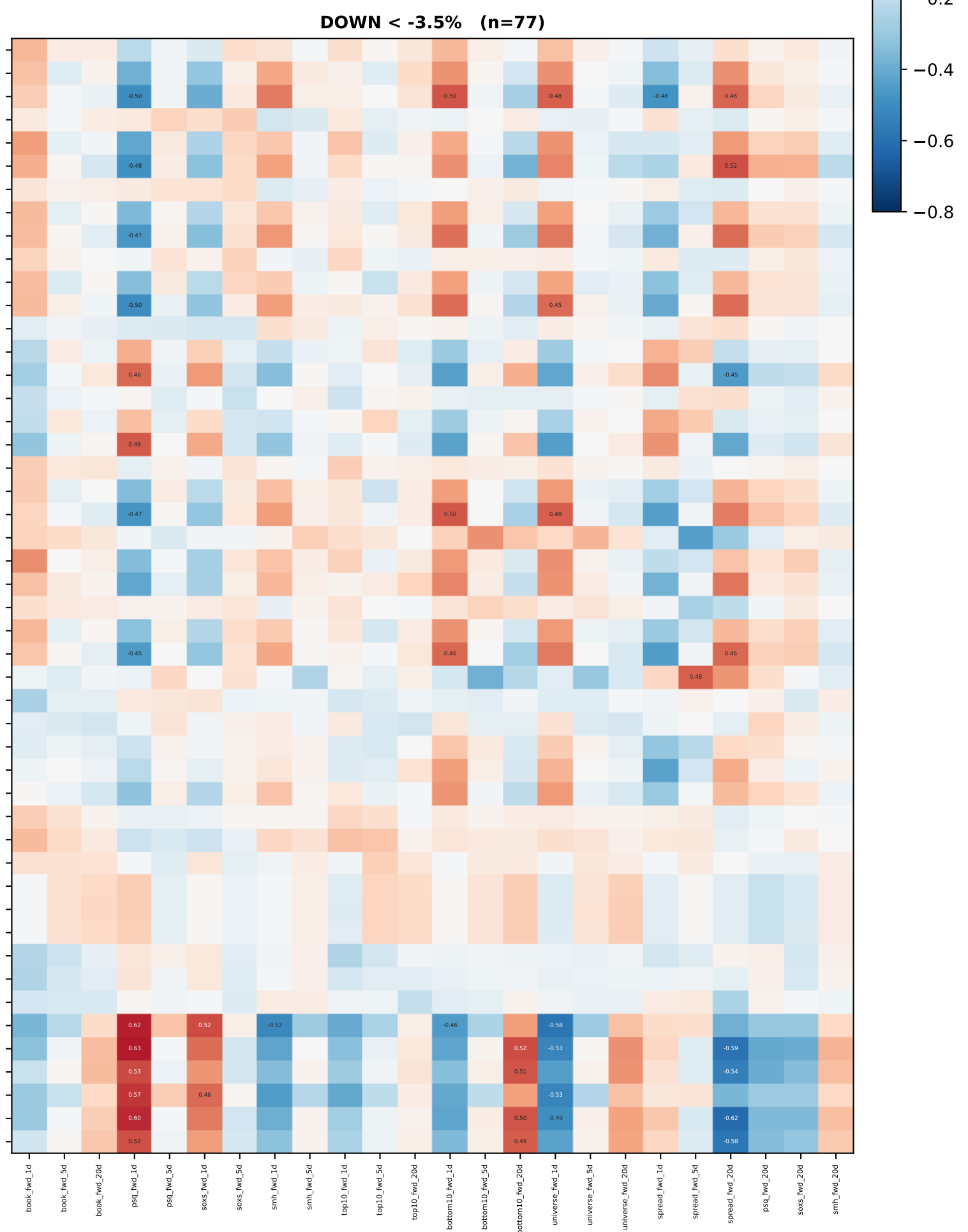
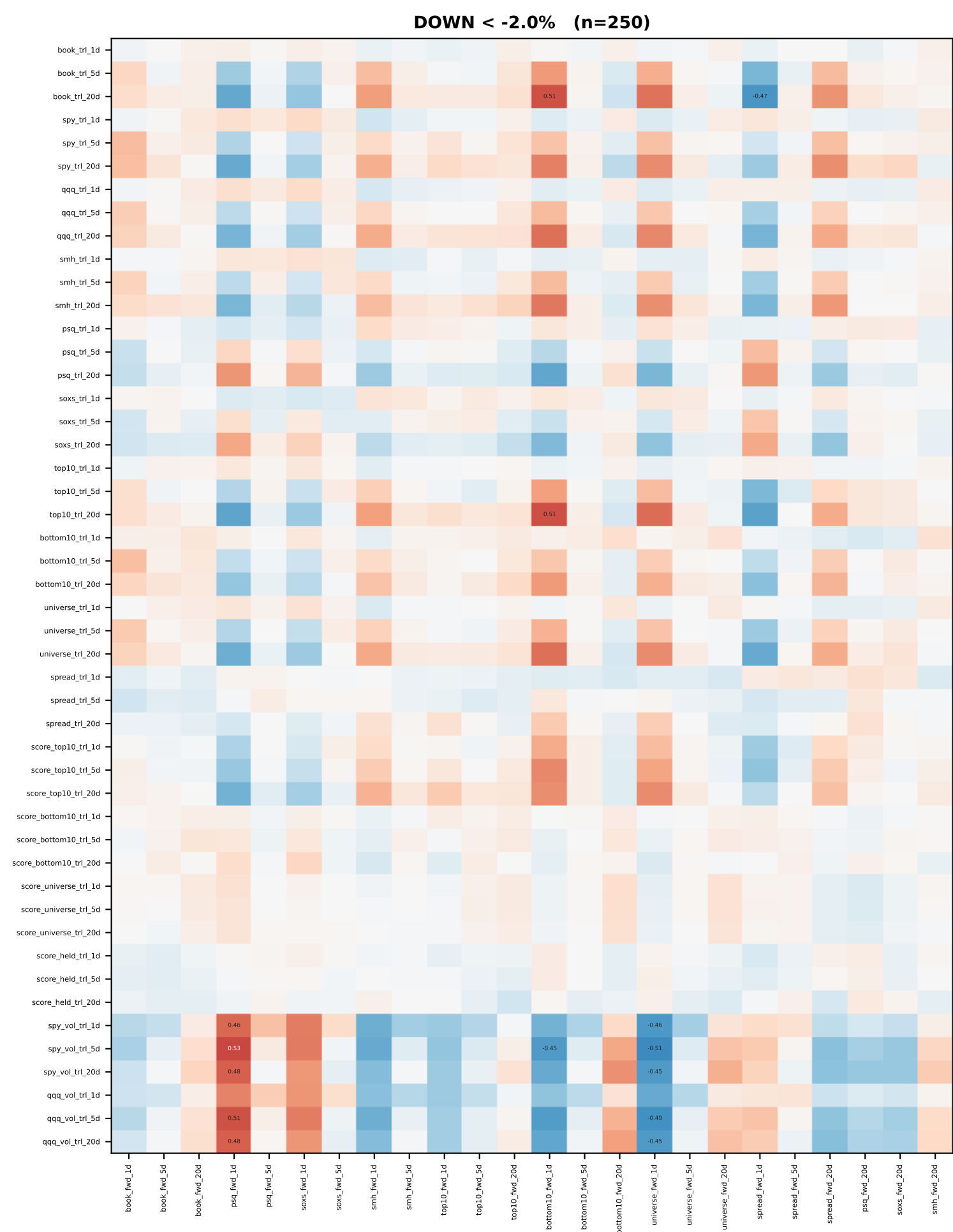
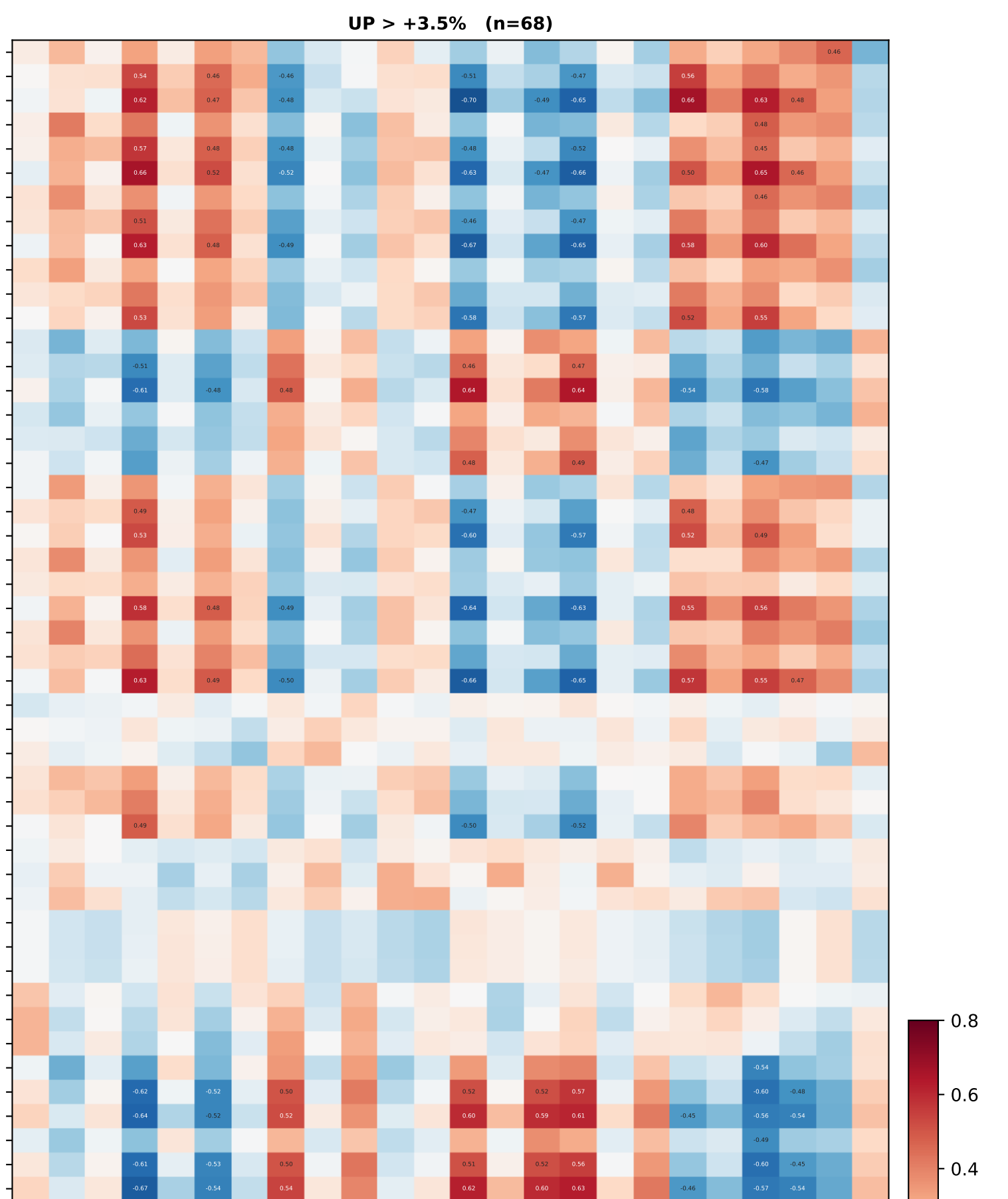
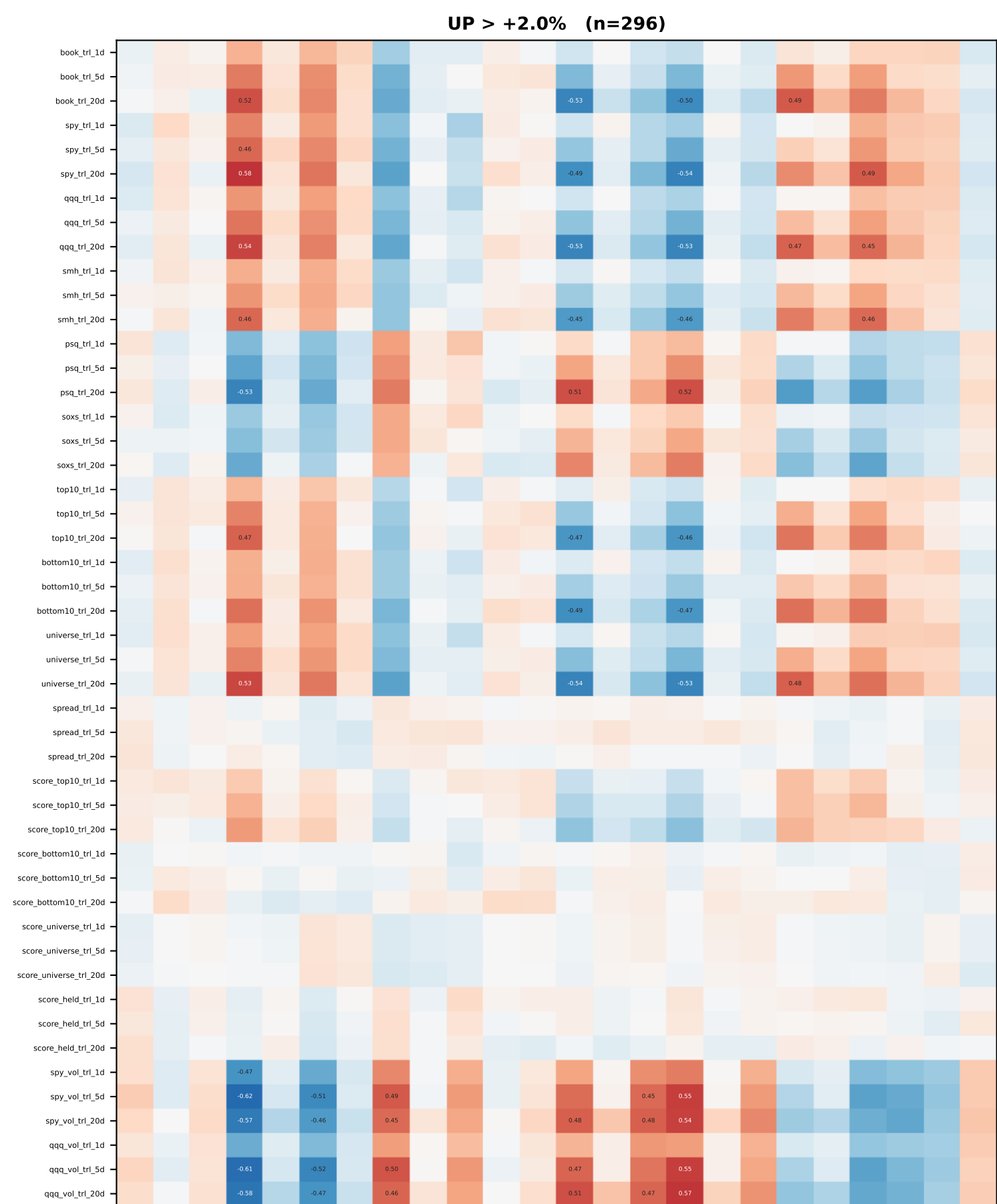
(none |r|>=0.5)

DOWN < -12.5% (n=70)

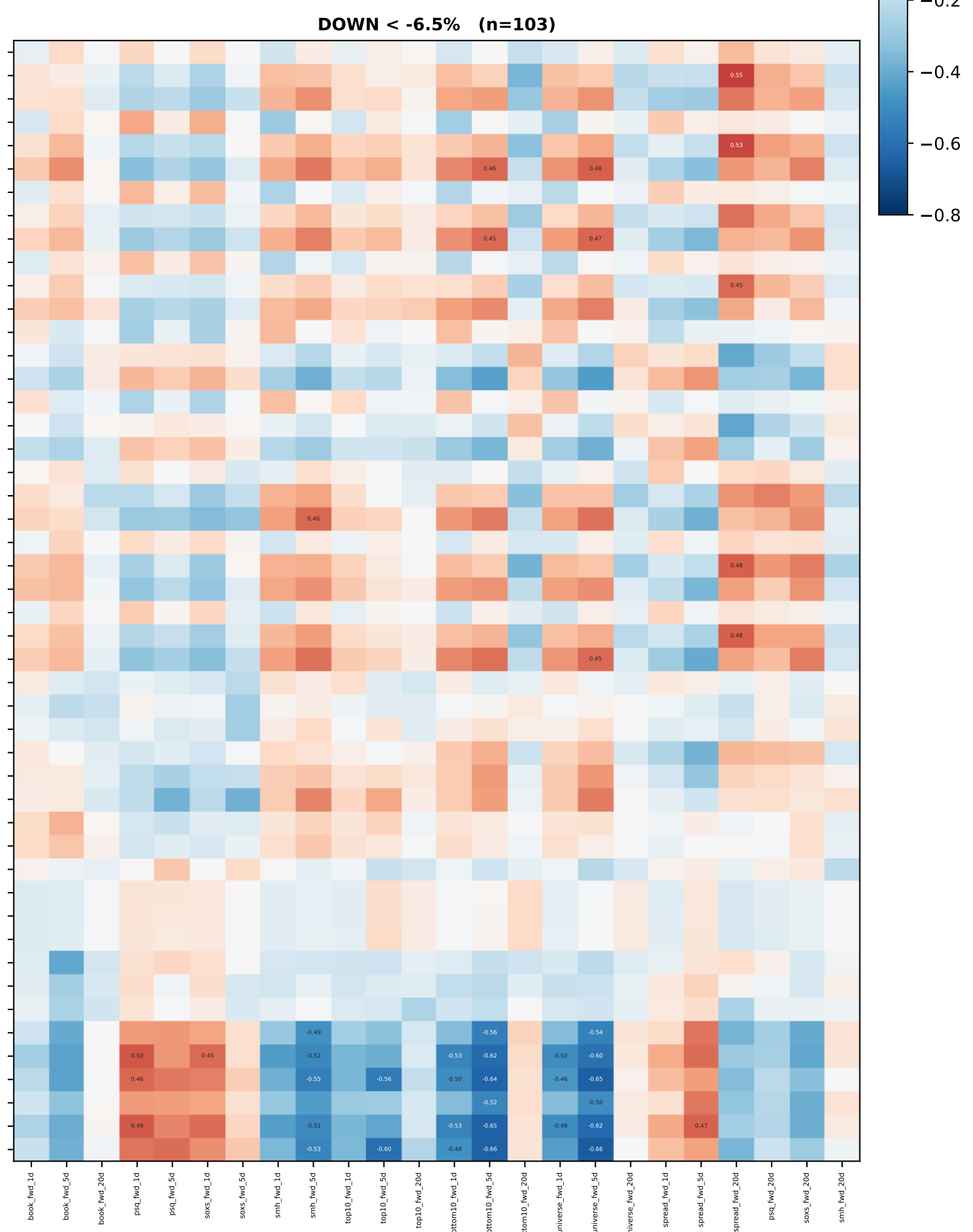
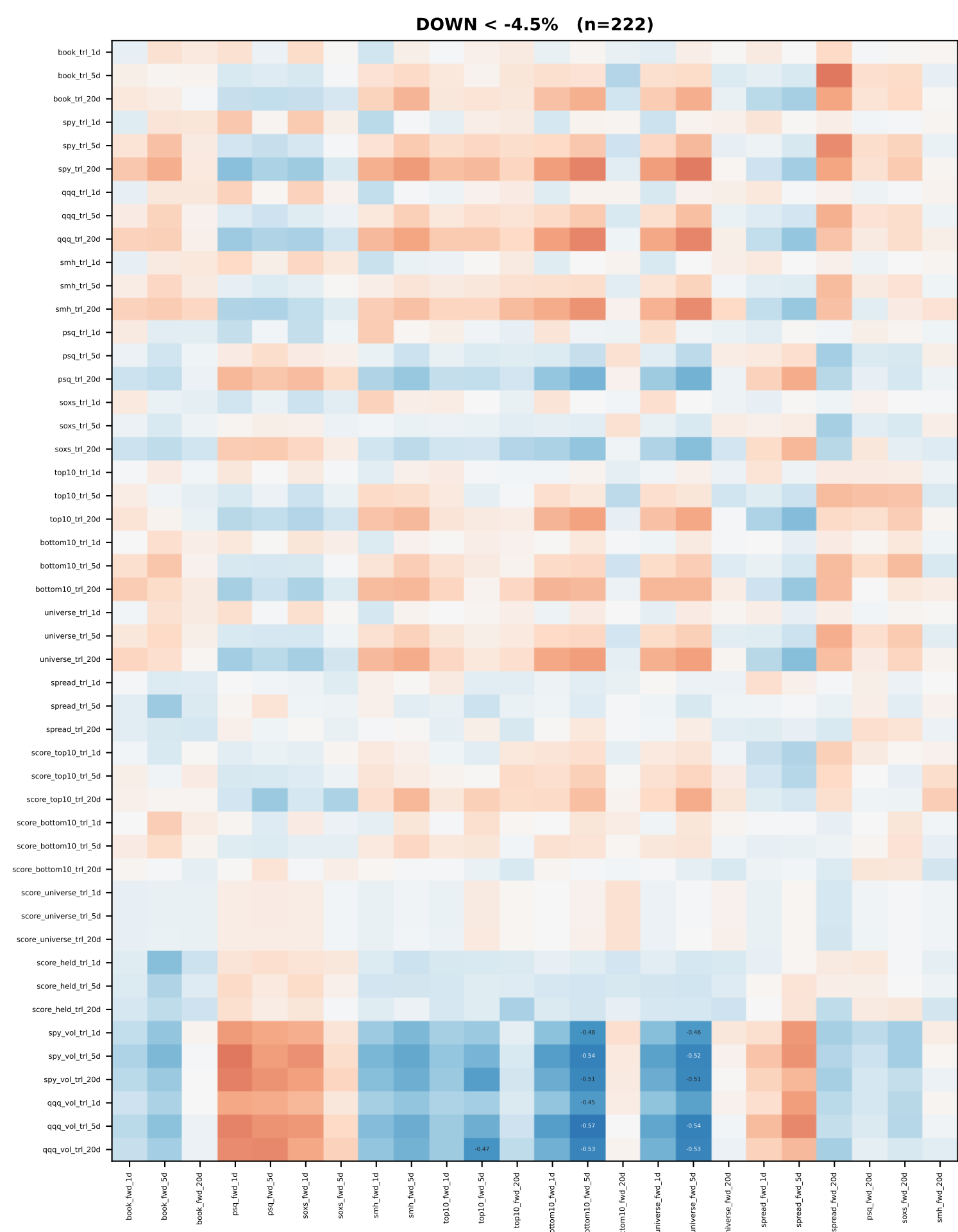
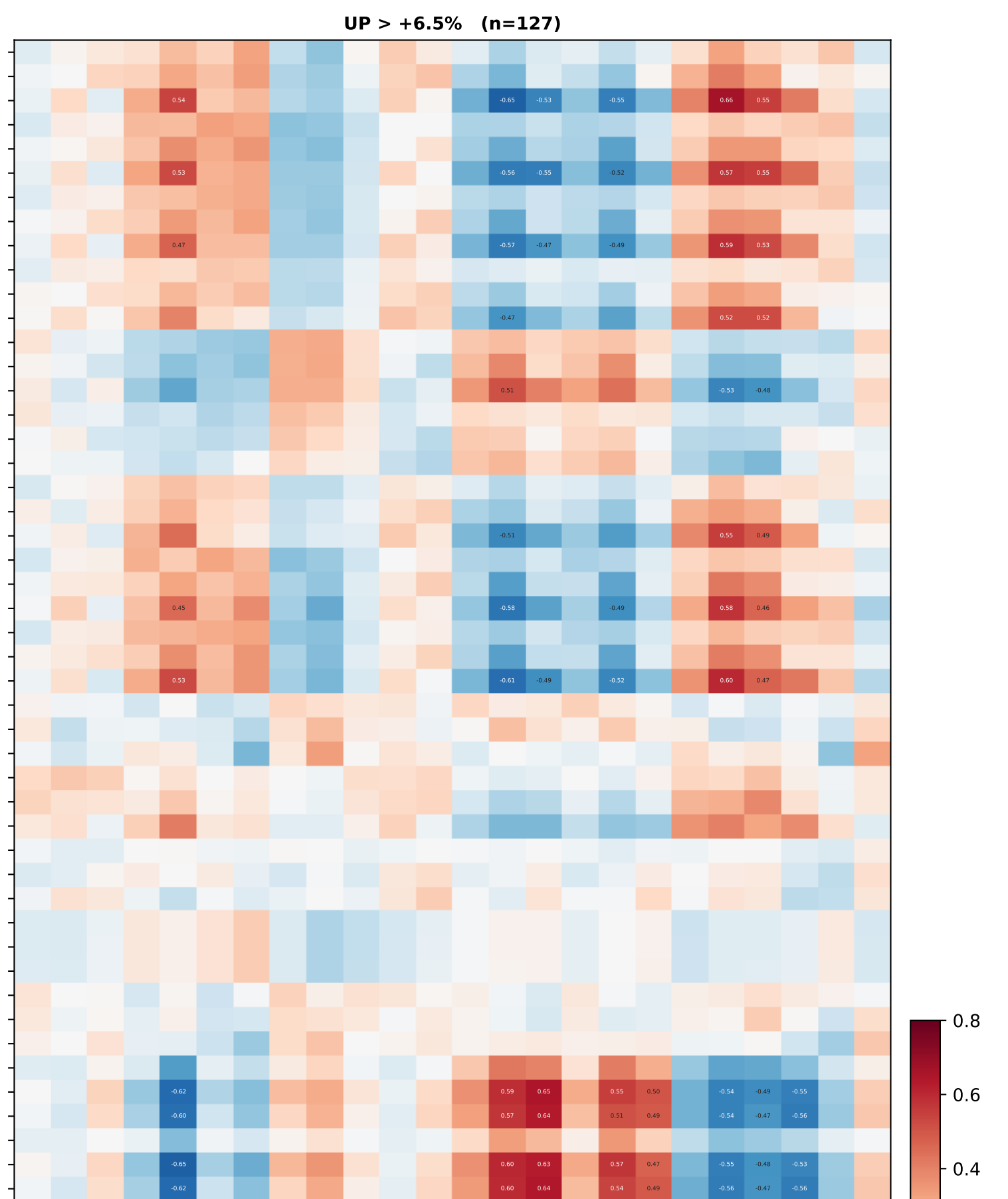
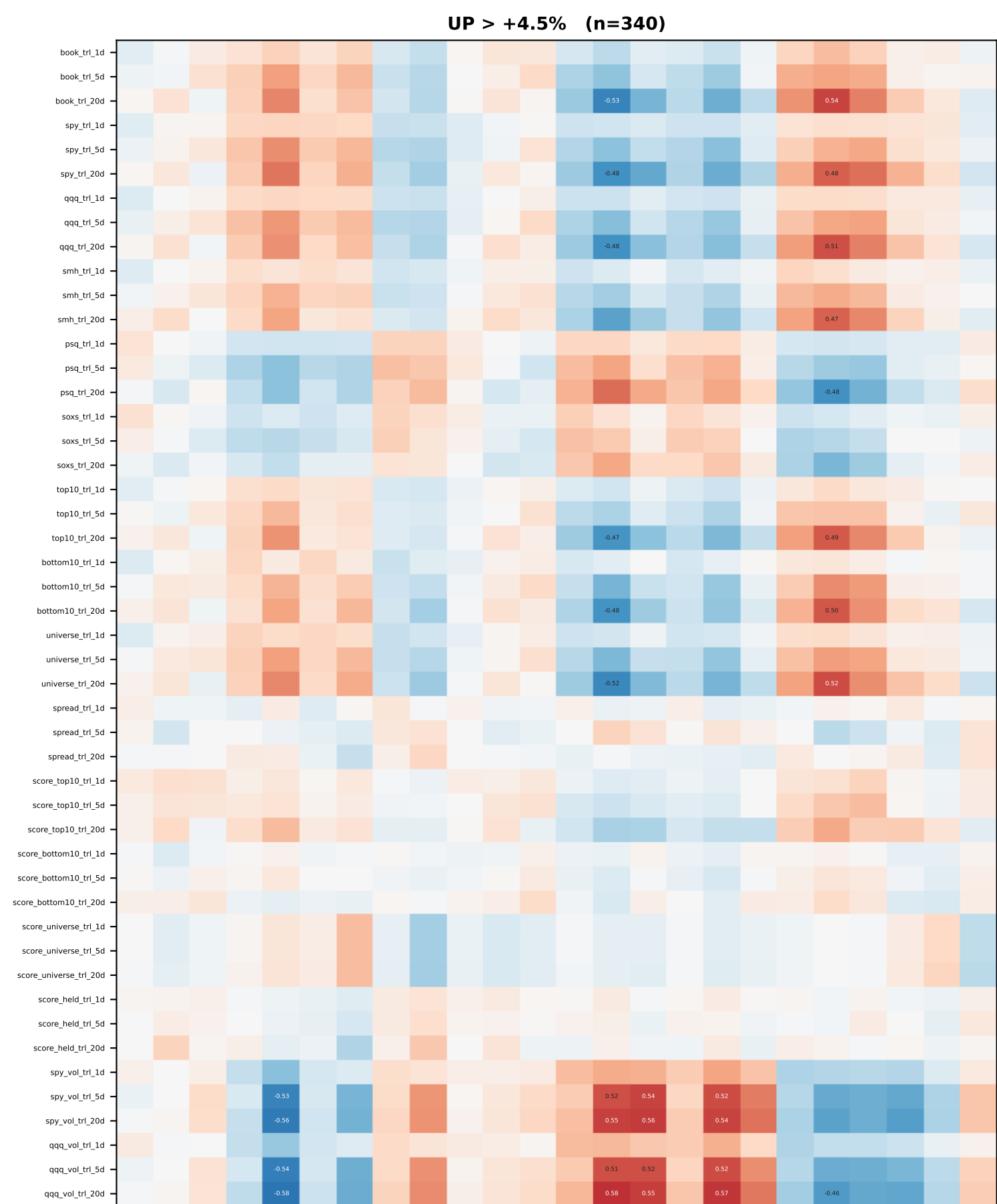
MODERATE (0.50-0.70):

-0.58	psq_trl_20d	avg=+0.1%	->	spread_fwd_20d	avg=-1.2%
-0.57	spy_vol_trl_5d	avg=+15.7%	->	bottom10_fwd_5d	avg=-3.6%
+0.57	book_trl_20d	avg=-0.1%	->	spread_fwd_20d	avg=-1.2%
-0.57	spy_vol_trl_5d	avg=+15.7%	->	soxs_fwd_20d	avg=+35.9%
-0.56	qqq_vol_trl_5d	avg=+20.3%	->	soxs_fwd_20d	avg=+35.9%
+0.55	spy_trl_20d	avg=-0.2%	->	soxs_fwd_20d	avg=+35.9%
+0.55	qqq_trl_20d	avg=+0.3%	->	spread_fwd_20d	avg=-1.2%
-0.54	spy_vol_trl_20d	avg=+14.8%	->	bottom10_fwd_5d	avg=-3.6%
-0.53	qqq_vol_trl_20d	avg=+20.2%	->	soxs_fwd_20d	avg=+35.9%
-0.52	spy_vol_trl_20d	avg=+14.8%	->	soxs_fwd_20d	avg=+35.9%
-0.52	qqq_vol_trl_5d	avg=+20.3%	->	bottom10_fwd_5d	avg=-3.6%
+0.52	book_trl_5d	avg=+0.2%	->	spread_fwd_20d	avg=-1.2%
-0.51	score_top10_trl_1d	avg=0.831	->	book_fwd_20d	avg=-17.7%
-0.51	soxs_trl_20d	avg=+1.5%	->	spread_fwd_20d	avg=-1.2%
+0.50	universe_trl_20d	avg=+0.6%	->	spread_fwd_20d	avg=-1.2%

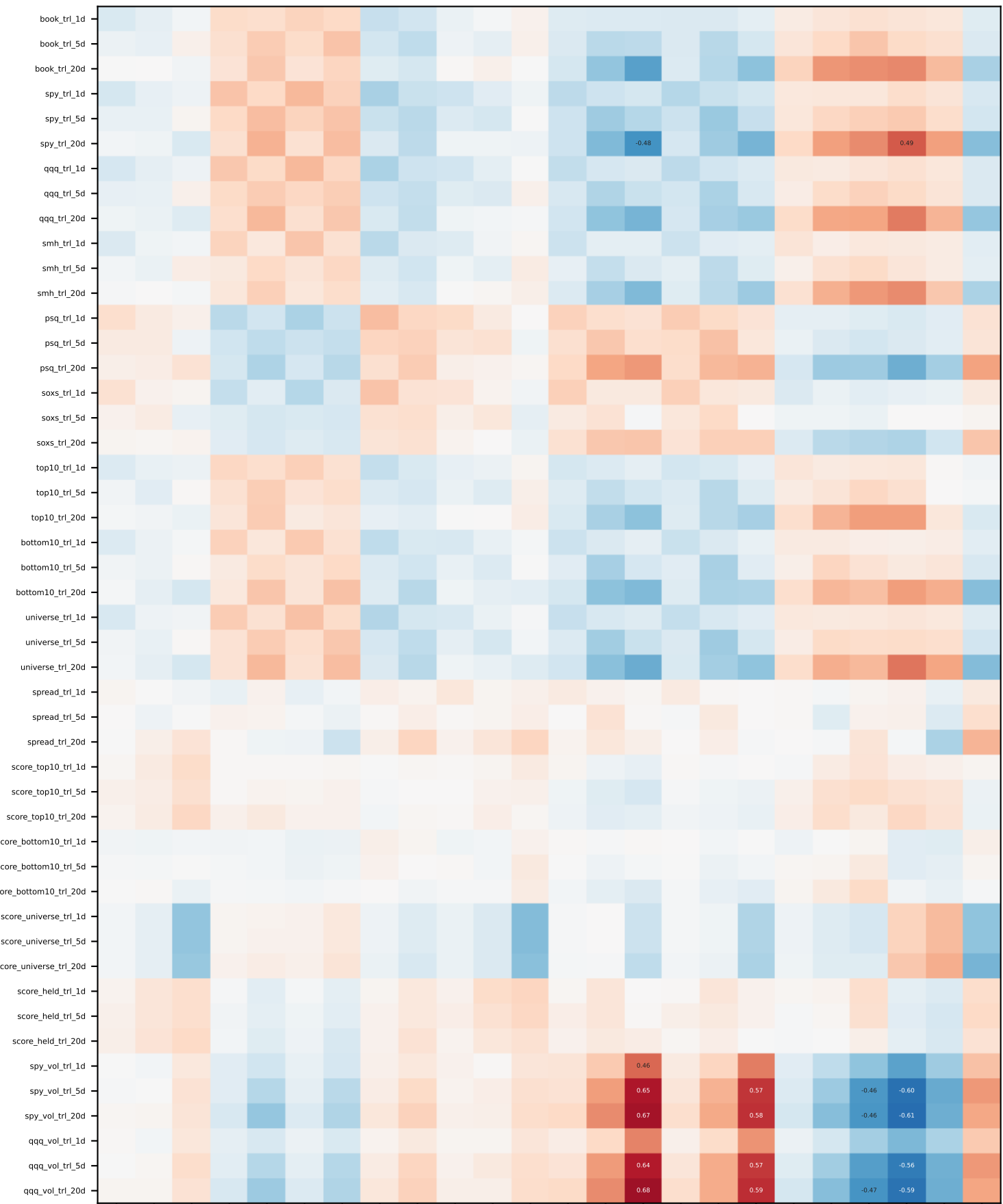
Trailing (rows) vs Forward (cols) correlations — filtered by Forward 1-Day portfolio return | scale +/-0.8



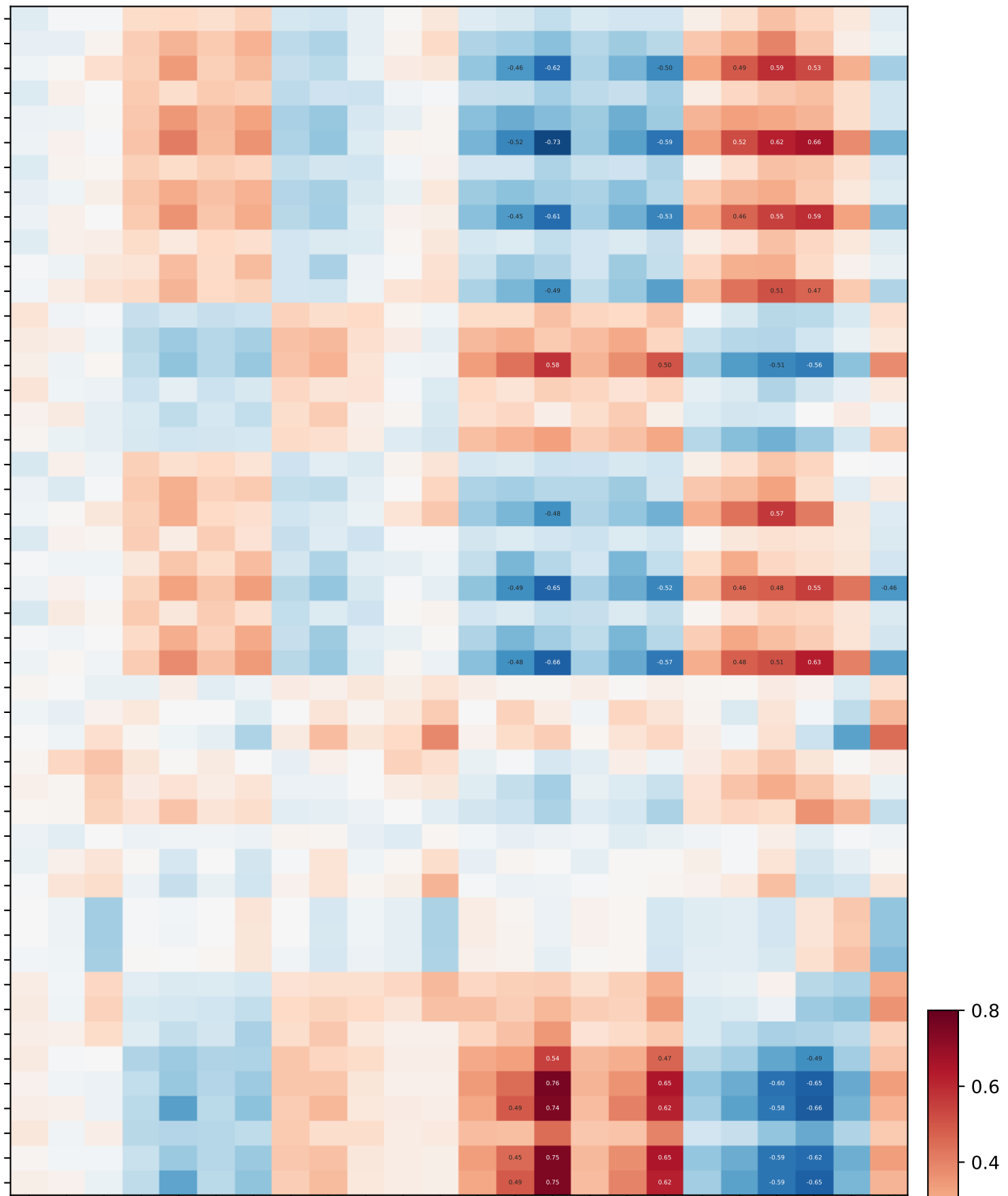
Trailing (rows) vs Forward (cols) correlations — filtered by Forward 5-Day portfolio return | scale +/-0.8



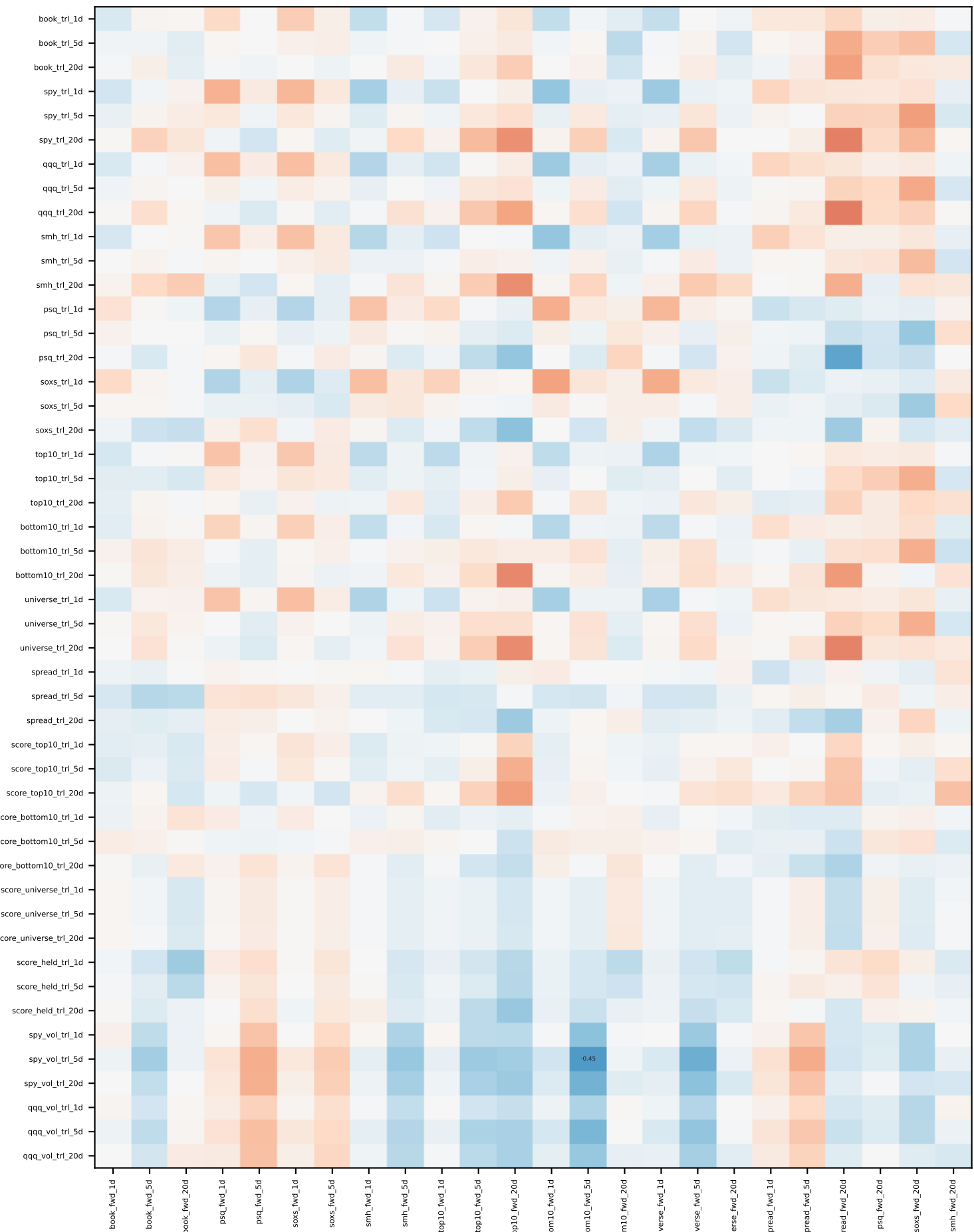
UP > +7.5% (n=655)



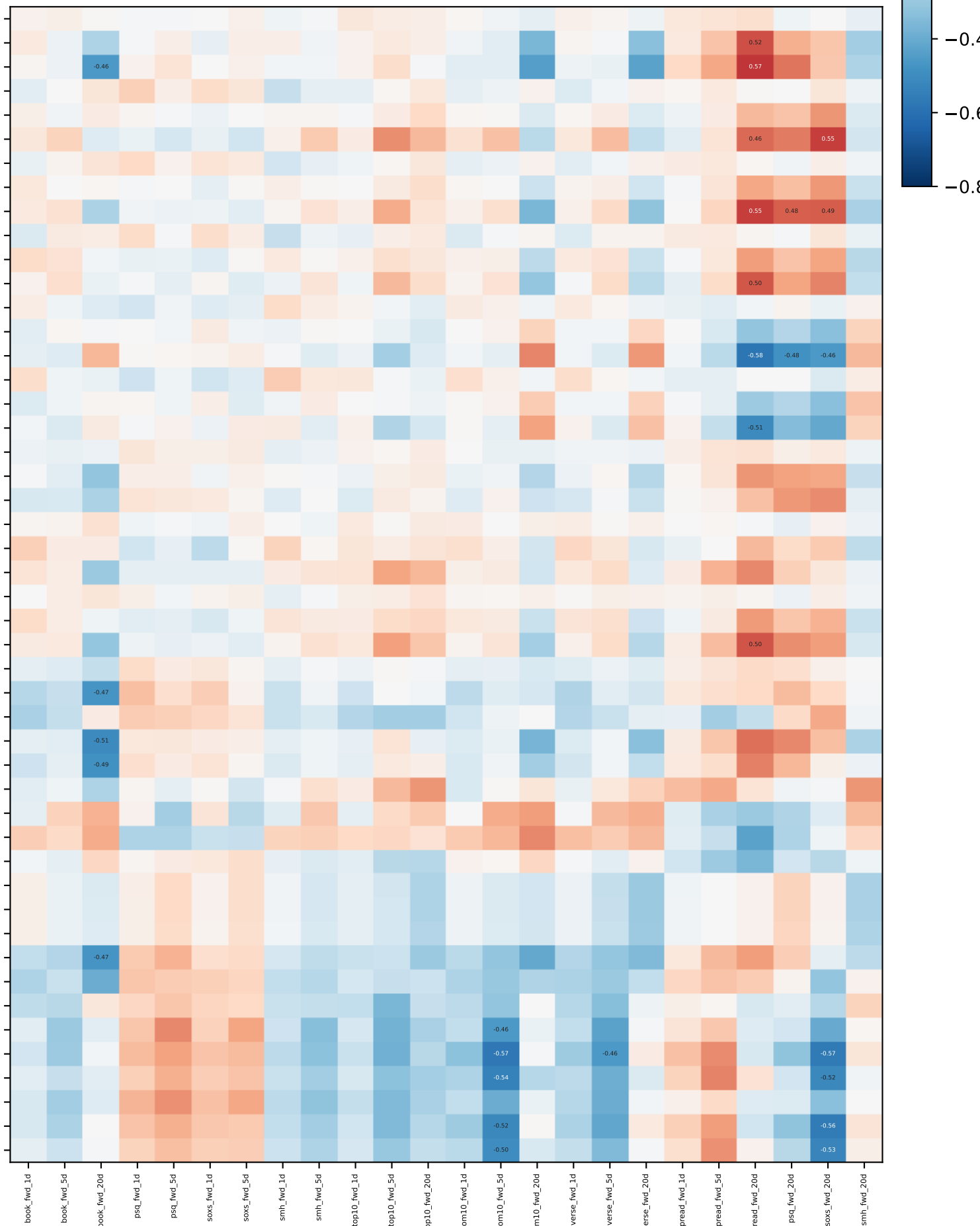
UP > +12.5% (n=268)



DOWN < -7.5% (n=215)



DOWN < -12.5% (n=70)



Pearson r